G2-Computer Science-Pseudocode & Python Worksheet

1. Write a Python program that uses a count-controlled loop (`for` loop) to ask the user to enter 5 test marks (integers). The program must calculate and output the total and the average mark.

2. Write a Python program using a conditional loop (`while` loop) that repeatedly asks the user to enter a positive integer.
   - Entering `-1` serves as the rogue value (sentinel) to stop the loop.
   - Any other negative number or zero should display an error message "`Invalid input, must be positive or -1 to stop`" and not be processed.
   - After the user enters -1, output the count of valid numbers entered.

3. A teacher wants to find the highest score achieved in an exam. Write a Python program that:
   - Prompts the teacher to enter how many student scores will be recorded.
   - Uses a loop to input each score.
   - Finds and prints both the highest score and lowest score entered.

4. Write a Python program that initializes a 1-D array (list) containing 6 test scores: `[45, 62, 78, 55, 90, 68]`.

   Your program must:

   - Use a count-controlled loop (`for` loop) to iterate through the array.
   - Calculate the total sum of all the marks in the array.
   - Calculate and display the average mark.

5. A weather station records the daily temperature for one week using a 1-D array:

   ```
   temperatures = [28, 31, 29, 32, 30, 33, 27]
   ```

   Write a Python program that:

- Loops through the array.
- Count how many days recorded a temperature strictly higher than 30°C.
- Outputs the total count with an appropriate message.

Linear search

6. Write a Python program that pre-populates a 1-D array with 5 student names:

```
names = ["Ali", "Sarah", "Dev", "Mei", "Carlos"]
```

The program must:

- Ask the user to enter a name to search for.
- Use a loop and a boolean flag (`found = False`) to perform a linear search through the array.
- If the name is present, display `"Found at position "` followed by its index position (0 to 4).
- If the name is not in the array after checking every element, display `"Name not found"`

Bubble Sort

Question 7

Write a Pseudocode and Python program that uses the **Bubble Sort** algorithm to sort a 1-D array of 5 integers in ascending order (smallest to largest).

Pre-populated array: `numbers = [42, 15, 8, 23, 4]`

Your program must:

- Use nested loops to compare adjacent elements.
- Use a temporary variable (`temp`) to swap elements if they are in the wrong order.
- Print the sorted array after sorting is complete.

Question 8

Write a Pseudocode and Python program that sorts a list of 6 student test scores in **descending order** (highest to lowest) using an optimized Bubble Sort.
Pre-populated array: `scores = [65, 88, 72, 95, 60, 81]`
Your program must:

- Sort the list from highest to lowest.
- Include a boolean flag (e.g., `swapped`) to detect if any swaps occurred during a pass.
- Terminate the sorting process early using a conditional loop (`while`) if a complete pass is made with no swaps (meaning the list is already sorted).
- Output the final sorted scores and the total number of passes it took to sort.

Question 9

Write a Pseudocode and Python program that uses a subroutine (procedure) named `CalculateTotal` which takes an array of 5 integers as a parameter, uses a count-controlled loop to calculate the sum of all elements, and outputs the result.

Question 10

Write a Pseudocode and Python program that uses a subroutine (function) named `GetValidScore`. The function must:

- Accept two integer parameters: `MinScore` and `MaxScore`.
- Use a conditional loop (`WHILE` or `REPEAT...UNTIL`) to prompt the user to input an integer between `MinScore` and `MaxScore` inclusive.
- Output an error message and repeat the prompt if the input is outside this range.
- Return the valid score to the calling program.

The main program should call this function twice to get scores for Paper 1 (range 0 to 75) and Paper 2 (range 0 to 50), then calculate and output the total score.

End of Question Paper